Advanced Data Engineering
03: Data Acquisition and Preparation

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Agenda

- Data ingestion/collection
- Data annotation/labelling
- Data preprocessing:
  - Data Cleaning
  - Data Transformation
  - Data Integration
  - Data Resample
- Feature engineering:
  - Categorical data encoding
  - Feature creation
  - Feature extraction
  - Feature selection
  - Dimensionality reduction

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Data format

Type	Recommended format	Not recommended
Text documents	• Unicode text (.txt) • PDF/A (.pdf) • ODT (.odt)	• Microsoft word (.doc) • Office Open XML (.docx) • Rich Text File (.rtf)
Spreadsheet	• CSV (.csv) • ODS (.ods)	• Microsoft Excel (.xls) • Office Open XML (.xlsx) • PDF/A (.pdf)
Database	• SQL (.sql) • CSV (.csv)	• Microsoft Access (.mdb) • HDF5 (.hdf5, .h5)
GIS	• GeoJSON (.json)	• Esri Shapefiles (.shp) • Esri Geodatabase (.gdb)
Images	• JPEG (.jpg, .jpeg) • TIFF (.tiff) • PNG (.png) • SVG (.svg)	• Adobe Illustrator (.ai) • WMF/EMF (.wmf, .imf) • CDR (.cdr)
Video	• Matroska (.mkv)	• MPEG-4 (.mp4) • MPEG-2 (.mpg) • AVI

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Data acquisition

- The process or activity of **observing** natural, social, or technical phenomena directly or indirectly **using certain equipment or tools** to produce data that can be **processed by a computer** to **answer certain questions**.
- Activities that are not acquiring data:
  - Collecting data by downloading datasets from data repositories on the Internet
  - Collecting data by purchasing technical datasets from data providers
  - Collecting data shared by colleagues
  - Collecting data randomly from the internet

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Requirements for data

Stages of identifying the context of needs

- Identify relevant stakeholders
- Acquire project & system documentation
- Document goals and objectives
- Summarize scope of capabilities
- Document impacts and constraints

Stages of interviewing stakeholders

- Identify candidates & review roles
- Develop interview questions
- Schedule & conduct interviews
- Summarize & identify gaps
- Resolve gaps & finalize results

Stages of requirements integration

- Document information workflow
- Identify required data elements
- Fact/Qualifier Matrix
- Harmonize data element semantics

Source and target mappings

- Propose target mappings
- Identify candidate data sources
- Source-to-target mapping

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Collecting data

- **Experiments**: conducted to test hypotheses in a structured environment, while field trials apply similar methods in real-world settings.
- **Surveys**: collect quantitative and qualitative data from a large number of respondents.
- **Interviews** and Focus Groups: direct, interactive sessions with individuals or small groups to collect in-depth insights.
- **Web** Scraping and **Social Media Monitoring**: automated techniques gather publicly available data from websites, social media platforms, etc.

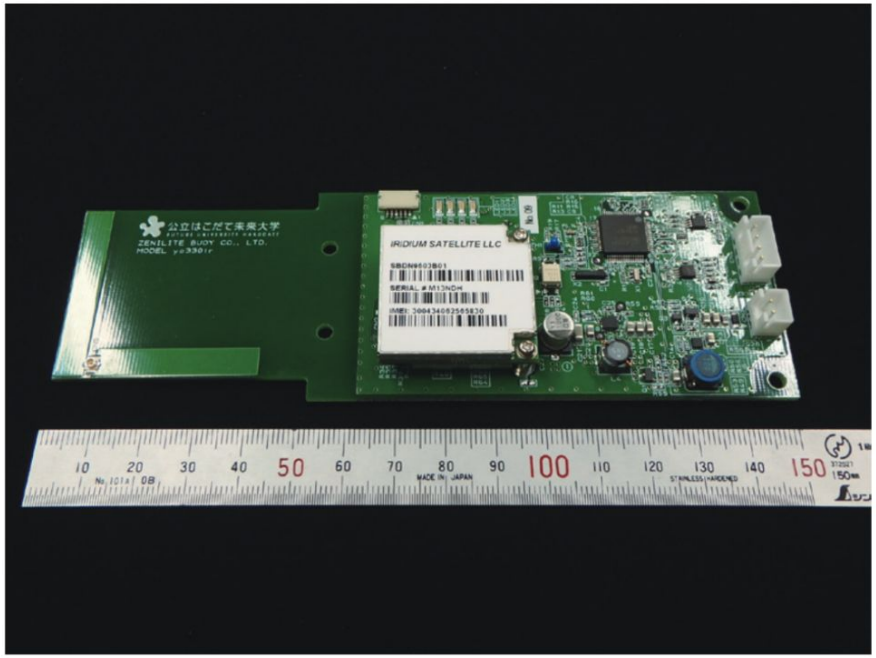
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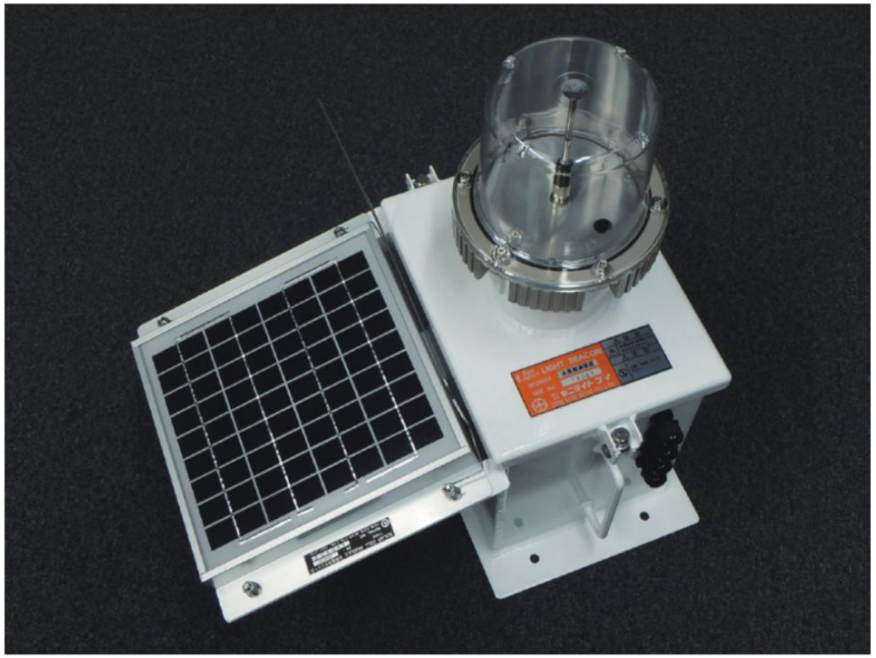
Sensor and IoT Data Collection



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Sensor and IoT Data Collection

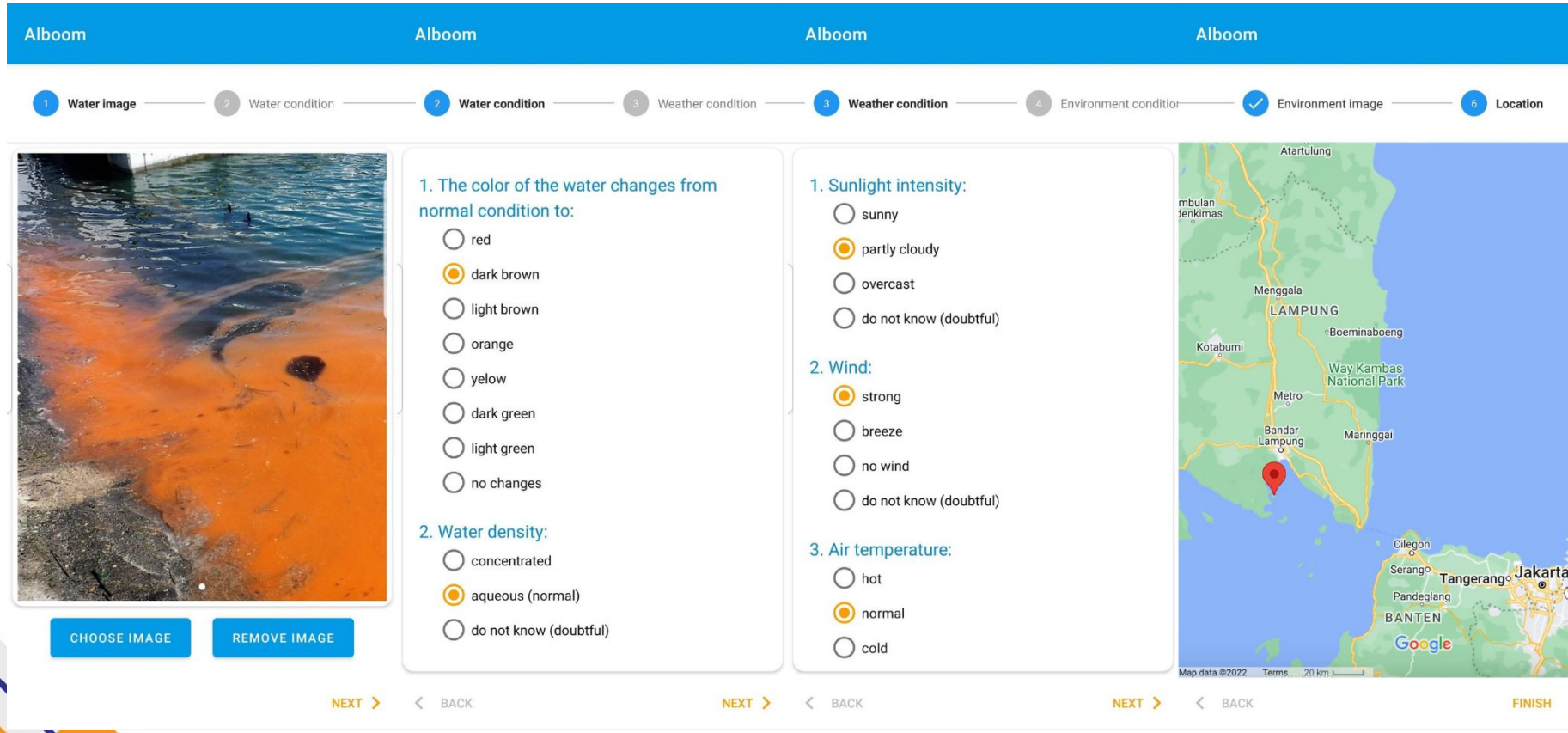




Although IoT sensors give accurate data about environmental conditions, these are expensive.

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Crowdsource-based data collection



https://doi.org/10.7717/peerj.14923

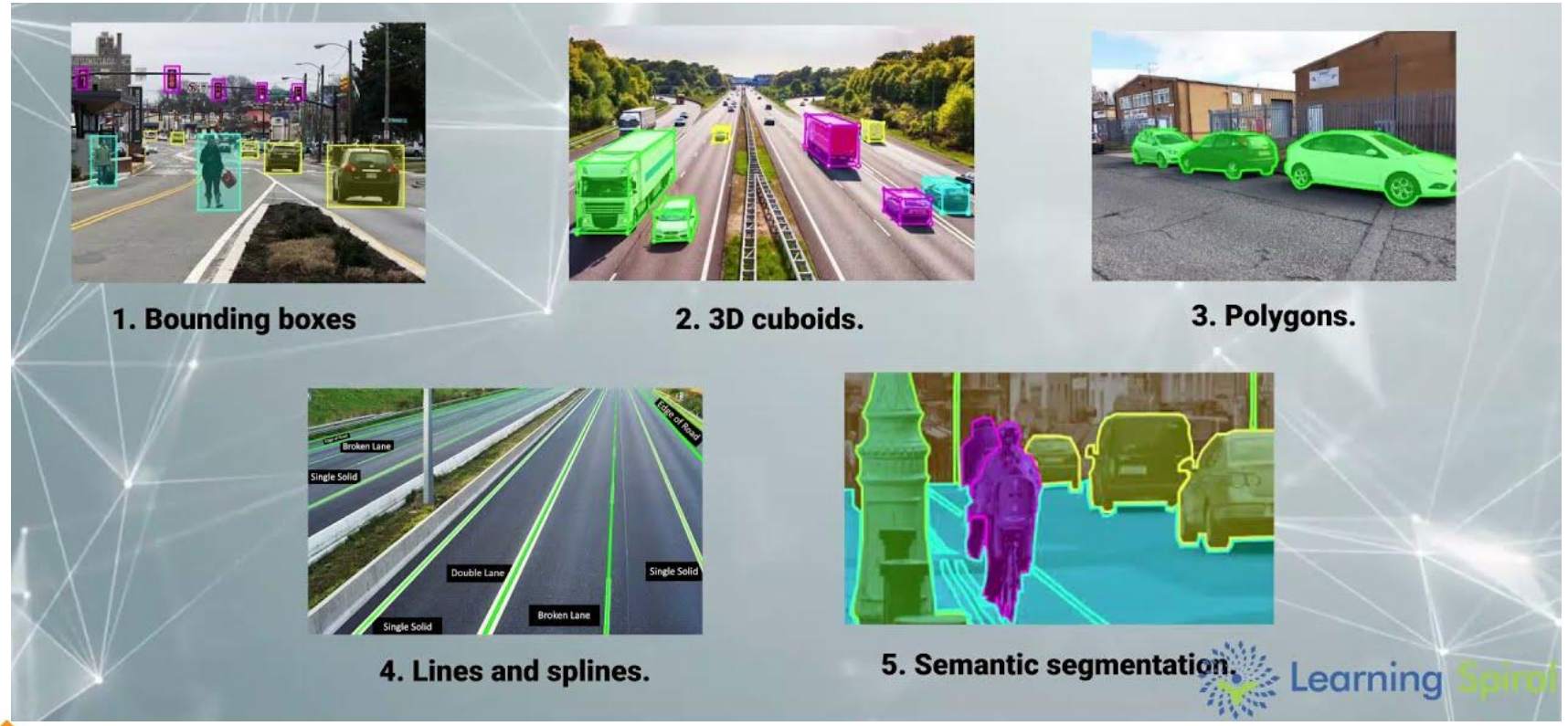
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Data annotation

- In supervised learning, data scientists/ ML engineers need labeled data to make machine (deep) learning models.
- While annotating data is not typically considered a core data engineering activity, data engineers may set up systems that **facilitate annotation process** since, normally, the annotators are domain experts not familiar with pipeline.
- Data engineers must **provide the necessary infrastructure** to handle and process the annotated data for ML applications.

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Data annotation for object detection



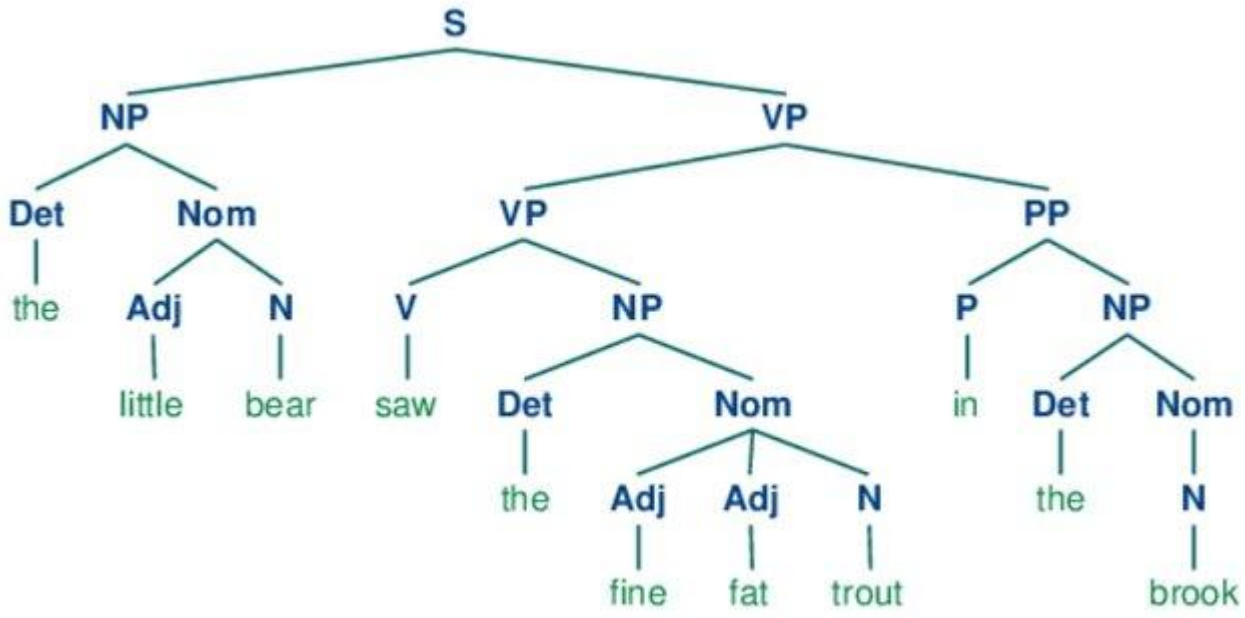
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Data annotation for entity recognition



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Data annotation for structure tagging



https://github.com/andryluthfi/annotation-tools

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Data preprocessing

- A critical step in the pipeline that transforms raw data into a clean and organized format, making it suitable for **analysis**, **modeling**, & **visualization**.
- Effective data preprocessing is essential for improving the **quality** of the dataset, which directly influences the **performance** and **accuracy** of the analysis or ML models.
- Systematically **cleaning**, **transforming**, **integrating**, and **resampling** data ensures the subsequent analytical processes are built on a solid foundation.

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Data cleaning

- Handling missing values:
  - **Deletion**: Remove rows or columns with too many missing values.
  - **Imputation**: Replace missing values with statistical measures like mean, median, or mode, or use advanced methods like K-nearest neighbors (KNN) imputation.
- Removing Duplicates:
  - Identify and remove duplicate records to ensure data uniqueness.
- Outlier Detection and Treatment:
  - Identify outliers using statistical methods (e.g., z-scores, IQR) and decide whether to remove, cap, or transform them to reduce impact.
- Noise Reduction:
  - Smooth out noisy data through techniques like binning, regression, or clustering methods.

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Data transformation

- Unit conversion:
  - Ensuring data is in consistent units.
- Normalization and Scaling:
  - **Normalization**: Rescale data to a [0, 1] range, which is especially useful for algorithms that require bounded input values.
  - **Standardization**: Transform data to have a mean of zero and a standard deviation of one.
- Logarithmic and Power Transformations:
  - Apply logarithmic transformation to reduce skewness in distributions, particularly for data that spans several orders of magnitude.
- Data Discretization and Binning:
  - **Discretization:** Convert continuous variables into categorical bins.
  - **Binning**: Group numerical values into bins to reduce noise and variability. This can be done using equal-width or equal-frequency

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Data integration and resampling

- Merging Data Sources:
  - Combine data from multiple sources or tables through joins or concatenation
- Handling Inconsistent Formats:
  - Resolve differences in data formats, units, or naming conventions to ensure a unified dataset.
- Resampling:
  - **Oversampling**: Involves increasing the number of instances in the minority class by duplicating existing data or creating synthetic samples (e.g., using SMOTE).
  - **Undersampling**: Involves reducing the number of instances in the majority class by removing some samples.

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Feature engineering

- Feature engineering is the process of transforming raw data into features that better represent the underlying problem, ultimately leading to improved performance for machine (deep) learning models:
- Effective feature engineering can dramatically enhance model performance by:
  - Revealing hidden relationships,
  - Reducing noise,
  - Lowering dimensionality,
  - And ensuring that features are presented in a form that makes sense to the model.

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Feature engineering

- Categorical data encoding:
  - **One-Hot Encoding**: Convert categorical variables into a binary vector format where each category is represented by a distinct column.
  - **Label Encoding**: Assign each unique category a numerical label.
  - **Frequency or Target Encoding**: Replace categorical values with their frequency in the dataset or with aggregated statistics (such as the mean target value), which can sometimes capture the inherent importance of each category.
- Feature creation:
  - **Derived Features:** Create new features by combining or transforming existing ones. For example, in a retail dataset, a "total purchase value" might be derived by multiplying quantity by unit price.
  - **Polynomial Features and Interactions:** Generate new features by taking powers or products of original features. This can capture non-linear relationships or interactions between variables that linear models might miss.

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Feature engineering

- Feature creation:
  - **Domain-Specific Transformations:** Apply transformations based on domain knowledge. In time series data, the day of the week, month, or seasonality indicators might be extracted.
- Feature extraction:
  - **Dimensionality Reduction**: Techniques such as Principal Component Analysis (PCA) or t-Distributed Stochastic Neighbor Embedding (t-SNE) are used to reduce the number of features while retaining most of the variance in the data. This is particularly useful when dealing with high-dimensional data.
  - **Autoencoders:** Use neural network architectures to learn a compressed representation of data. The hidden layer of an autoencoder can serve as a set of new, informative features.
  - **Text Vectorization:** Convert textual data into numerical features using methods like Term Frequency-Inverse Document Frequency (TF-IDF) or word embeddings (Word2Vec, GloVe, or BERT).

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Feature engineering

- Feature selection:
  - **Filter Methods:** Use statistical tests (e.g., chi-square, correlation coefficients) to select features based on their relevance to the target variable.
  - **Wrapper Methods**: Evaluate combinations of features by training a model and selecting the subset that yields the best performance (e.g., recursive feature elimination).
  - **Embedded Methods**: Incorporate feature selection as part of the model training process (e.g., Lasso regression, decision tree-based methods) that naturally penalize less important features.

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Task

- Create a Google Colab notebook. Demonstrate:
  1. Acquire countries dataset from `https://restcountries.com`
  2. Retrieve the latest currency exchange rates for each country relative to USD from other data sources and merge the data with the dataset
  3. Collect data of current weather conditions for capital cities for each country from other data sources and combine the data with the set
  4. Ingest COVID-19 data from `https://disease.sh` for each country and join the data with the dataset
  5. Perform data preprocessing and feature engineering:
     - Handle missing values
     - Encode categorical variables
     - Extract and scale features
     - Etc.
  6. Present final results of the dataset in a .csv file
  7. Visualize the data summarization

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